

House Price Prediction — Summary Report

Project Summary

This project develops an end-to-end supervised machine learning regression pipeline using the Ames Housing Dataset to predict residential house prices. The workflow includes exploratory data analysis, preprocessing, feature engineering, model training, evaluation, hyperparameter tuning, residual analysis, and deployment.

Business Problem

The objective is to build an automated house price estimation system that helps buyers, sellers, and real estate companies estimate fair property prices accurately.

Data Preprocessing & Feature Engineering

Missing values were handled using appropriate strategies such as filling categorical values with 'None', numerical values with median or zero where applicable, and dropping columns with excessive missing data. Outliers were removed, and new features including TotalSF, HouseAge, RemodAge, HasGarage, and HasPool were created. Categorical variables were encoded and numerical variables were scaled.

Models Implemented

Linear Regression, Ridge Regression, Lasso Regression, Random Forest Regressor, and XGBoost Regressor were trained and evaluated. Hyperparameter tuning was performed using RandomizedSearchCV.

Model Evaluation

Models were compared using RMSE, MAE, R^2 Score, 5-Fold Cross Validation RMSE, and training time. Ensemble methods generally achieved the best predictive performance.

Top Predictive Features

OverallQual, GrLivArea, TotalBsmtSF, GarageArea, GarageCars, YearBuilt, Neighborhood, and TotalSF were identified as the most influential features affecting house prices.

Business Interpretation

The model shows that construction quality, living area, basement size, garage capacity, neighborhood, and house age significantly influence selling prices. Such insights can support automated property valuation systems.

Conclusion

The project successfully demonstrates a complete supervised learning pipeline for house price prediction. After comparing multiple regression models, tree-based ensemble models such as Random Forest and XGBoost provided the strongest predictive performance and are suitable for practical deployment.